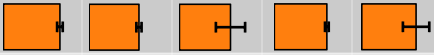
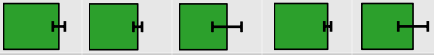
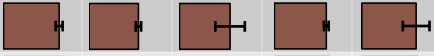
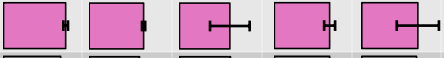
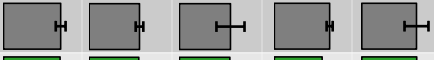
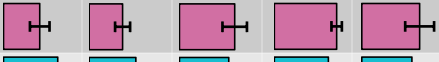
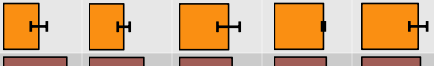
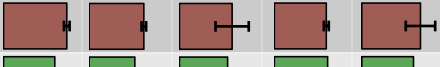
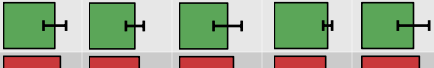
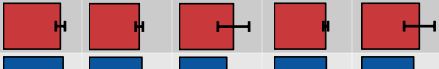
| Methods | Core Algorithm/Model | How to estimate kinetic parameters | | Whether to calculate and output du/dt | Performance at Cell level | | | | | Performance at Gene level | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Local | | Global | | | Gene-level kinetics | | | Kinetic parameters | |
| | | α | β&γ | | consistency score | ICCoh | CBDir | TransCosine | BATC | Real_CBDir | Sim_CBDir | Sim_CosSim | α_Spearman | γ_Spearman |
| scVelo_deterministic | Steady-state linear regression | —— | Gene-specific steady-state ratio (β/γ) | —— |  | | | | | —— | —— | —— | —— | —— |
| scVelo_stochastic | Stochastic model using second-order moments | —— | Gene-specific moment fit | —— |  | | | | | —— | —— | —— | —— | —— |
| scVelo_dynamic | Dynamical model with EM and analytical solution | Gene-specific | Gene-specific | Yes (model output) |  | | | | | | | | —— | —— |
| MultiVelo (RNA_only) | RNA-only ODE with chromatin held fixed | Gene-specific | Gene-specific | Yes (model output) |  | | | | | | | | —— | —— |
| UniTVelo | Unified-time dynamical model | Gene-specific RBF function | Gene-specific | Yes, not output |  | | | | | | | | —— | —— |
| UniTVelo(mode2) | Independent-time dynamical model | Gene-specific RBF function | Gene-specific | Yes, not output |  | | | | | —— | —— | —— | —— | —— |
| Dynamo | RNA velocity estimation + SparseVFC | Constant | β fixed γ gene-specific | —— |  | | | | | —— | —— | —— | | —— |
| TIvelo | Trajectory-guided velocity inference + orientation score | —— | —— | Explicit Output |  | | | | | | | | —— | —— |
| cell2fate | Bayesian linearized ODE in Pyro | Module-weighted production | Gene-specific | —— |  | | | | | —— | —— | —— | —— | —— |
| TFvelo | Gene regulatory network + generalized EM | TF-regulated production | —— | —— |  | | | | | —— | —— | —— | —— | —— |
| veloAE | Autoencoder representation learning | —— | β fixed latent-space γ | —— |  | | | | | —— | —— | —— | —— | —— |
| veloVI | Deep generative model (VAE) | Latent state | Gene-specific | Yes (latent velocity output) |  | | | | | | | | —— | —— |
| DeepKINET | Deep generative kinetic model | —— | Cell-specific (VAE Output) | Yes (model output) |  | | | | | —— | —— | —— | —— | |
| TSvelo | Neural ordinary differential equation | Continuous function α(t) | Gene-specific | Yes (model output) |  | | | | | —— | —— | —— | | —— |
| cellDancer | Deep neural network | Cell-specific | Cell-specific (DNN Output) | Yes (model output) |  | | | | | | | | | |
| DeepVelo | Graph convolutional network | Cell-specific | Cell-specific (GCN Output) | Yes (model output) |  | | | | | | | | | |
| SDEvelo | Stochastic differential equation | Sigmoid function | Gene-specific | —— |  | | | | | —— | —— | —— | —— | —— |
| DeepVelo_scv | Neural ODE + VAE reconstruction | —— | —— | from scVelo |  | | | | | | | | —— | —— |
| GraphVelo_scv | Tangent-space projection + kNN velocity correction | —— | —— | from scVelo |  | | | | | | | | —— | —— |
| GraphVelo_dyn | Tangent-space projection + kNN velocity correction | —— | —— | —— |  | | | | | —— | —— | —— | —— | —— |